Certification, Realization, and Obstruction:

A Universal Fiber Architecture

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Abstract

This paper closes the editorial loop for a multi-manuscript program in machine-checked mathematics. The executive claim is that a single *collapse / refinement / fiber* architecture—certification at a bare carrier, strict comparison upstairs, obstruction organized as fiber and section data—is instantiated internally in dependent type theory, validated on twelve external mathematical families, and then discharged into algebra, arithmetic (embedding problems), topology (Quillen for Galois connections), and logic/computability (self-certification and halting anchors). We state the role of each companion paper, a cross-domain dictionary, route labels used in the program (C/A/B/D), and what remains open. The intended reader is an advanced mathematician who has not read the series; citations point to the formal manuscripts and the Lean repository.

Trust boundary: This manuscript is a program map and interpretive synthesis; definitive statements remain in the cited formal papers and Lean modules.

Suite context. The Infinity Compression paper suite is organized as follows: the *flagship manuscript* states the *internal origin theorem*; the *transferability paper* validates the same packaging on twelve external benchmarks; *general-method manuscripts* (group extensions; Quillen) supply theorem-level leverage as the *algebraic and arithmetic engine* and *topology discharge*; the *synthesis manuscript* maps the full program; the *summit manuscript* distills reflective non-exhaustion; the *technical Mathlib companion* is reviewer-facing. This manuscript is the *synthesis map*.

Keywords: formal verification, certification vs. realization, fibers, obstruction, Infinity Compression, Lean 4, program architecture

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Code/Data availability: Repository: https://github.com/novaspivack/infinity-compression-lean. Root module `InfinityCompression.lean`; build via `lake build`. Toolchain pin in `lakefile.lean`; see `MANIFEST.md` at repository root.

1 Premises, disagreement coordinates, and trust boundary

Where disagreement can (and cannot) live

Formal spine. Program-level routing and "what compresses" claims summarize formal outputs; they do not replace verification of individual theorem files.

Premise coordinates (for external readers).

- **Synthesis vs. lemma files.** Editorial synthesis may streamline dependencies; proof obligations live in the cited papers. *Logical effect:* Disputing the map is not, by itself, a refutation of a specific formal lemma.
- **Open frontiers.** Explicit open problems in the series are stated in the flagship and summit manuscripts, not erased here. *Logical effect:* This map does not claim a strongest possible inevitability theorem over all frameworks.

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2 Executive statement

One-sentence thesis. Canonical certification at a collapsed bare carrier does not exhaust structurally significant information: the gap is organized by a forgetful map, nontrivial fibers, and explicit obstructions—and that pattern repeats across internal formalization, external benchmarks, and classical mathematical domains.

What compresses. Quotient language, section language, obstruction language, and the implementation-vs.-certificate distinction are often treated as separate methodological topics. This program shows they are instances of one *comparison architecture*: fix carriers, fix a forgetful map π, ask where collapse occurs, and analyze fibers and obstructions when it does not. That compression—from scattered idioms to one reusable pattern—is the synthesis-level payoff.

3 Certification, realization, and non-exhaustion

Certification. *Certification* means a theorem-level invariant packaged as a canonical record at a *bare* carrier (proof-irrelevant at that carrier when the type forces it): many derivation routes may exist, but the official statement names a single certified outcome. It does *not* mean informal "approval" or social certification; it means the library's designated bare certificate type and its collapse theorems.

Realization. *Realization* means structured or proof-relevant data above that bare layer: witnesses, roles, sections, splittings, nerve data, or fiber points that remain mathematically load-bearing once $\pi : E \to B$ is fixed. It is not mere "extra syntax": it is the carrier on which non-injectivity and fiber lemmas become meaningful.

Non-exhaustion. *Non-exhaustion* is the proved claim that certification does not determine realization: distinct legitimate data can share the same bare image under π, organized by fibers and obstruction laws. It does *not* assert that every formal system exhibits this phenomenon; it is established on *named* carriers and explicit hypotheses in the artifact.

Examples (suite instances). Flagship Infinity Compression: collapse at the bare summit record; non-injectivity of π_A on enriched splits [1]. Group extensions: splitting iff trivial cocycle [3]. Benchmark families: structured bundles beyond flat baselines [2]. Quillen / Galois connections: contractible fibers feeding Theorem A data [4].

4 Flagship internal theorem

The Infinity Compression flagship paper [1] is the *internal origin theorem*: collapse of standard extraction, reflective split, strict refinement of the typed forgetful map, and partial fiber classification—all in the library's native carriers.

5 External validation (transferability)

The twelve-tranche survey [2] is the *transferability paper*: the same schema applies to quotients, products, dependent sums, coproducts, subtypes, orbit and ideal quotients, pushouts, classifiers,

localization, pullbacks, and a non-surjective control. Its job is not theorem-level leverage in classical fields; it rules out idiosyncrasy.

6 Algebraic and arithmetic engine

The group-extension paper [3] is the *algebraic and arithmetic engine*: section cocycles, splitting iff trivial cocycle, embedding problems as extension splitting, local-to-global obstruction, and the documented bridge toward Galois-theoretic towers.

7 Topological discharge

Quillen's Theorem A for Galois connections [4] is the *topology discharge*: nerves, contractible fibers over down-sets, and homotopy-equivalence data at the simplicial level for the Galois-connection case.

8 Logic and computability anchor

Route D in the program supplies a computability-facing discharge: self-certification phenomena aligned with standard undecidability infrastructure (e.g. halting and Rice scaffolding in the Mathlib `ComputablePred` layer), making the analogy to classical incompleteness precise without identifying the main theorem with Gödel's syntax. Exact statements live in the Lean `RouteD/` and `Summit/` modules; this synthesis paper does not restate every definition.

9 Cross-domain dictionary

The following table is a navigational index, not a formal commuting diagram.

Domain	**Bare / cert.**	**Enriched**	**Comparison**	**Headline / obstruction**
IC flagship	canonical bare cert.	enriched split	π_A	non-inj.; nontrivial fiber
T1–T12	flat baseline	structured bundle	projection	fibers, sections (T12: boundary)
Group ext.	section exists	splitting data	section data	split $\Leftrightarrow$ trivial cocycle
Embedding	quotient data	liftings	extension cmp.	solvable $\Leftrightarrow$ trivial cocycle
Quillen GC	nerve image	homotopy witn.	nerve maps	contractible fibers
Computability	cert. layer	realized behav.	cert. cmp.	halting / Rice

10 Routes C, A, B, and D (program labels)

- **Route C (headline):** reflective non-exhaustion at the flagship carriers.
- **Route A (infrastructure):** supporting general lemmas (e.g. collision patterns, forgetful infrastructure) used by the headline.
- **Route B (applications):** algebra, topology, arithmetic bridges—the general-method papers.
- **Route D (logic/computability):** halting and Rice-style scaffolding aligned with computability theory rather than informal metaphor.

11 Why this is a reusable research instrument

The architecture is portable because it isolates *where* collapse occurs, *where* comparison must move upstairs, and *what* invariant obstructs lifting—independent of any single application domain. Lean provides a checkable artifact; the paper suite separates internal origin, transferability, domain discharges, and summit-level claims.

Operational recipe.

1. Identify the bare certificate carrier B.
2. Identify an enriched carrier E tracking realization data.
3. Define the forgetful or comparison map $\pi : E \to B$.
4. Prove or locate *collapse*: when many routes hit one bare point in B.
5. Test injectivity of π; if it fails, work on fibers $\pi^{-1}(b)$.
6. Extract obstruction laws (cocycle, contractibility, non-certifiability) governing liftability and equivalence.

12 What this architecture is not

- **Not** a claim that all of mathematics reduces to one master theorem.
- **Not** a generic restatement of dependent-type proof relevance: the contribution is *where* collapse and non-injectivity are proved, on named carriers.
- **Not** mere packaging rhetoric: the suite contains formal theorems with zero `sorry` in the scoped modules.
- **Not** (yet) a single universal quantified Lean theorem over all domains at once: that unification remains a frontier goal.

13 Portability vs. inevitability

The current suite establishes strong *portability*: the same comparison pattern reappears across internal, benchmark, and classical discharges. A stronger *inevitability* claim—that every reflective certification setup must exhibit non-exhaustion in a single abstract class—remains a summit-level ambition, not a finished theorem.

14 What remains open

Full fiber classification in the flagship setting; complete simplicial homotopy formalization behind Quillen; full H^1/H^2 classification in Mathlib; stronger unified quantification across all carriers in a single declaration; publication and Mathlib submission scheduling.

Acknowledgments

Repository: https://github.com/novaspivack/infinity-compression-lean.

References

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